

INDR262

SPRING 2026 TERM PROJECT

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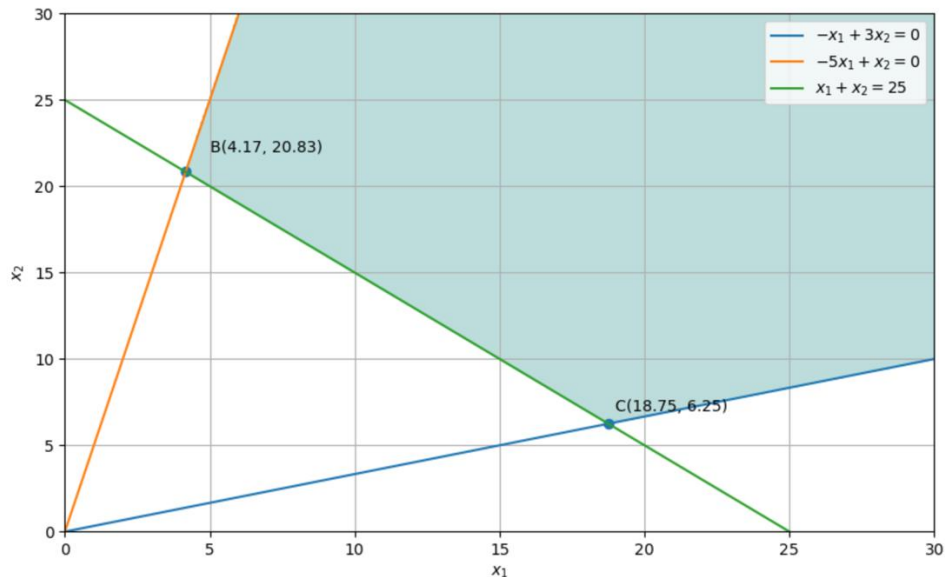
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Question 1:

1.a- 1.b)

The feasible region is the area located above the green line, above the blue line, and below the orange line in the graph. The intersection point of the constraints are:



$$(0,0), (0,25), (25,0), \left(\frac{75}{4}, \frac{25}{4}\right), \left(\frac{25}{6}, \frac{125}{6}\right)$$

But between these points, only these points are in the feasible region:

$$\left(\frac{75}{4}, \frac{25}{4}\right), \left(\frac{25}{6}, \frac{125}{6}\right)$$

For these points, objective function values are:

$$Z\left(\frac{75}{4}, \frac{25}{4}\right) = 425$$

$$Z\left(\frac{25}{6}, \frac{125}{6}\right) = 483.33$$

Since the purpose is reaching the minimum value, optimal solution is:

$$x_1^* = \frac{75}{4}, x_2^* = \frac{25}{4}$$

And optimal value is:

$$Z^* = 425$$

1.c)

$$\text{Minimize } z = 16x_1 + 20x_2$$

$$\text{Subject to: } -x_1 + 3x_2 \geq 0$$

$$x_1 - 3x_2 + x_3 = 0$$

$$-5x_1 + x_2 \leq 0$$

$$-5x_1 + x_2 + x_4 = 0$$

$$x_1 + x_2 \geq 25$$

$$x_1 + x_2 - x_5 + r_1 = 25$$

$$x_1, x_2 \geq 0$$

$$x_1, x_2, x_3, x_4, x_5, r_1 \geq 0$$

Phase I: *Minimize* $Z = r_1$

BV	Z	x_1	x_2	x_3	x_4	x_5	r_1	RHS
z	1	0	0	0	0	0	-1	0
x_3	0	1	-3	1	0	0	0	0
x_4	0	-5	1	0	1	0	0	0
r_1	0	1	1	0	0	-1	1	25

BV	Z	x_1	x_2	x_3	x_4	x_5	r_1	RHS
z	1	1	1	0	0	-1	0	25
x_3	0	1	-3	1	0	0	0	0
x_4	0	-5	1	0	1	0	0	0
r_1	0	1	1	0	0	-1	1	25

Entering $\rightarrow x_1$ Leaving $\rightarrow x_3$

BV	Z	x_1	x_2	x_3	x_4	x_5	r_1	RHS
z	1	0	4	-1	0	-1	0	25
x_1	0	1	-3	1	0	0	0	0
x_4	0	0	-14	5	1	0	0	0
r_1	0	0	4	-1	0	-1	1	25

Entering $\rightarrow x_2$ Leaving $\rightarrow x_4$

BV	Z	x_1	x_2	x_3	x_4	x_5	r_1	RHS
z	1	0	0	0	0	0	-1	0
x_1	0	1	0	1/4	0	-3/4	3/4	75/4
x_4	0	0	0	3/2	1	-7/2	7/2	175/2
x_2	0	0	1	-1/4	0	-1/4	1/4	25/4

Final Tableau

1.d)

In this model slack variables are defined as x_3 and x_4 . Surplus variable is x_5 and r_1 states the artificial variable. To ensure computational convenience first constraint is reformulated by multiplying -1.

In Phase I method, aim is obtaining initial feasible solution by minimizing the artificial variable. Therefore, Phase I objective function is structured as:

$$\min Z = r_1$$

because of simplex iteration, r_1 variable is removed from basis. When the artificial variable leaves the basis, its value becomes zero.

$$r_1 = 0$$

Thus, at the end of Phase I that objective value is obtained:

$$Z^* = 0$$

the objective function being equal to zero shows that model is feasible. Therefore, initial feasible solution is achieved for original problem, and we conclude that Phase II step can be applied.

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----- PHASE I SOLVER RESULT -----
Phase I Objective Value W*: 0.00
x1 = 18.750
x2 = 6.250
x3 = 0.000
x4 = 87.500
x5 = 0.000
x6 = 0.000
x7 = 0.000
The original problem is feasible.
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1.e)

$$\text{Minimize } z = 16x_1 + 20x_2$$

Subject to:

$$x_1 - 3x_2 \leq 0$$

$$-5x_1 + x_2 \leq 0$$

$$\text{Minimize } z = 16x_1 + 20x_2 + Mr_1$$

Subject to:

$$x_1 - 3x_2 + x_3 = 0$$

$$-5x_1 + x_2 + x_4 = 0$$

$$x_1 + x_2 \geq 25$$

$$x_1 + x_2 - x_5 + r_1 = 25$$

$$x_1, x_2 \geq 0$$

$$x_1, x_2, x_3, x_4, x_5, r_1 \geq 0$$

BV	Z	x_1	x_2	x_3	x_4	x_5	r_1	RHS
Z	1	-16	-20	0	0	0	-10000	0
x_3	0	1	-3	1	0	0	0	0
x_4	0	-5	1	0	1	0	0	0
r_1	0	1	1	0	0	-1	1	25

BV	Z	x_1	x_2	x_3	x_4	x_5	r_1	RHS
Z	1	9984	9980	0	0	-10000	0	250000
x_3	0	1	-3	1	0	0	0	0
x_4	0	-5	1	0	1	0	0	0
r_1	0	1	1	0	0	-1	1	25

Entering -> x_1

Leaving -> x_3

BV	Z	x_1	x_2	x_3	x_4	x_5	r_1	RHS
Z	1	0	39932	-9984	0	-10000	0	250000
x_1	0	1	-3	1	0	0	0	0
x_4	0	0	-14	5	1	0	0	0
r_1	0	0	4	-1	0	-1	1	25

Entering -> x_2

Leaving -> r_1

BV	Z	x_1	x_2	x_3	x_4	x_5	r_1	RHS
Z	1	0	0	-1	0	-17	-9983	425
x_1	0	1	0	1/4	0	-3/4	3/4	75/4
x_4	0	0	0	3/2	1	-7/2	7/2	175/2
x_2	0	0	1	-1/4	0	-1/4	1/4	25/4

Final tableau

----- BIG-M RESULT -----

Objective Value: 425.00

x1 = 18.750

x2 = 6.250

x3 = 0.000

x4 = 87.500

x5 = 0.000

x6 = 0.000

x7 = 0.000

Question 2:

2.a)

Objective Function:

$$\min Z = \sum_{p \in P} \sum_{c \in C} \sum_{r \in R} c_{pr} y_{pcr}$$

The goal here is to minimize the total cost of assigning proctors. The formula takes the cost of putting a proctor in a specific room (c_{pr}) and multiplies it by our decision variable (y_{pcr}), which tracks whether that proctor-room assignment actually happens.

Constraints:

$$\sum_{r \in R} x_{cr} = 1 \quad \forall c \in C$$

This constraint means that each course must be assigned to exactly one room.

$$\sum_{c \in C, t_c = t} x_{cr} \leq 1 \quad \forall r \in R, \forall t \in T$$

This constraint means that one room can have at most one exam in the same time slot.

$$s_c x_{cr} \leq K_r \quad \forall c \in C, r \in R$$

This constraint means that the room must have enough capacity for the students of the course.

$$\sum_{p \in P} y_{pcr} = q_c x_{cr} \quad \forall c \in C, r \in R$$

This constraint means that if a course is assigned to a room, it must get the required number of proctors.

$$\sum_{\substack{c \in C: \\ t_c = t}} \sum_{r \in R} y_{pcr} \leq a_{pt} \quad \forall p \in P, t \in T$$

This constraint means that a proctor can work in a time slot only if they are available.

$$\sum_{\substack{c \in C: \\ t_c = t}} \sum_{r \in R} y_{pcr} \leq 1 \quad \forall p \in P, t \in T$$

This constraint means that a proctor can supervise at most one exam in the same time slot.

$$x_{cr} \in \{0,1\} \quad \forall c \in C, r \in R$$

This means that x_{cr} is a binary variable. It is 1 if course c is assigned to room r , and 0 otherwise.

$$y_{pcr} \in \{0,1\} \quad \forall p \in P, c \in C, r \in R$$

This means that y_{pcr} is a binary variable. It is 1 if proctor p is assigned to course c in room r , and 0 otherwise.

2.b)

We solved the exam scheduling model with binary decision variables. This means that each variable can take only 0 or 1. The objective of the model is to minimize the total proctor assignment cost. The model also makes sure that each course is assigned to exactly one room, the room capacity is enough, proctors are available, and a proctor does not supervise more than one exam in the same time slot.

After solving the model with the solver, the optimal objective value is:

$$Z^* = 189750$$

This means that the minimum total proctoring cost is **189750**. Therefore, this is the optimal cost for the binary exam scheduling model.

Part 2b Total Cost: 189750.00

2.c)

In this minimization problem the optimal solution of continuous model is same with binary model's optimal solution. Optimal value of binary model is:

$$Z_{IP} = 189750$$

The optimal value of continuous model is:

$$Z_{LP} = 189750$$

Part 2c Objective: 189750.00

This means that the LP relaxation does not improve the objective value. The reason is that the binary model already reaches the lowest possible cost.

Continuous model can reach better solution than Binary model solution. However, the continuous model is not fully realistic in real life. In the continuous model, variables can take values like 0.5. This may mean assigning half of a proctor or half of a room to an exam. In real life, this is not possible. A proctor is either assigned or not assigned, and a course is either assigned to a room or not assigned. Because of this, the binary model is more suitable for the real exam scheduling problem.

2.d)

TA073 Exam-Day Schedule:

TA073 is assigned to exam INDR209 in time slot T6, room ENG109.

TA073 is assigned to exam INDR269 in time slot T1, room SOS101.

Door Schedule for SCI107:

Room SCI107: exam INDR249, time slot T3

Room SCI107: exam INDR282, time slot T4

Room SCI107: exam INDR285, time slot T6

Room SCI107: exam INDR301, time slot T2

Room SCI107: exam INDR313, time slot T5

2.e)

The disciplinary constraints evolve four TA: TA018, TA055, TA040, and TA065. The constraint states that there is no pair among these four TA's assigned to the same room for same exam. The C(4,2) combination gives 6 possible pairs. So, the following constraints are (p1, p2) and for every course $c \in C$ and room $r \in R$:

$$y_{\{p1,c,r\}} + y_{\{p2,c,r\}} \leq 1$$

For all $(p1, p2) \in \{(TA018, TA055), (TA018, TA040), (TA018, TA065), (TA055, TA040), (TA055, TA065), (TA040, TA065)\}$, $\forall c \in C, \forall r \in R$.

After implying this constraints and solving this model, the optimal cost is still \$189750 which means cost increase is 0 relative to the part b.

In conclusion, the new constraints do not change the total cost. However, because large scheduling models often have multiple optimal solutions, the same total cost does not mean the exact same proctor assignments are used. Without checking every decision variable, we cannot say for sure that the original plan remained unchanged. We can only conclude that the system found a solution with the same minimum cost.

PART 2e RESULT

Part 2b Objective Value: 189750.00

Part 2e New Objective Value: 189750.00

Increase Relative to Part 2b: 0.00

2.f)

In the solution, some assignment decisions were predetermined in advance. These fixed assignments are added as additional constraints in model, and it reduces the solver's flexibility in making assignment decisions. Thus, the model cannot evaluate all possible assignment options flexibly.

As a result of decreasing flexibility, we ensure bigger objective value in the minimization problem. Previous optimal result is:

$$Z_{old} = 189750$$

At the end of bonus scenario obtained objective value is:

$$Z_{new} = 200732$$

Thus, total planning cost is increased by:

$$\$200732 - \$189768 = \$10982$$

Part 2b Total Cost: 189750.00
-----PART 2f RESULT-----
Part 2b Objective Value: 189750.00
Part 2f New Objective Value: 200732.00
Increase Relative to Part 2b: 10982.00

Question 3:

3.a)

There are 50 items listed in Knapsack262.csv file. All items have mass(kg), volume(m³), and value(dollar).

Decision Variable:

$x_i = 1$ if item i is selected,

$x_i = 0$ otherwise

Parameters:

d_i = value of item i

m_i = mass of item i

v_i = volume of item i

We are trying to maximize the value from items that we can fit in the knapsack. Our objective function will be:

- Maximize $\sum_{i=1}^{50} d_i x_i$

Since we have mass' and volume's upper bound we need to add two constraints:

Subject to:

- $\sum v_i x_i \leq 1262$ (Volume Constraint)
- $\sum m_i x_i \leq 1262$ (Mass Constraint)

Also the question asks for binary decision variables we need to add variable constraints:

- $x_i \in \{0, 1\} \forall i$

3.b)

We solved the same knapsack problem under three different constraints: binary, integer and continuous decision variables. The objective function, volume and mass constraints are identical in the three models. Since the domain of decision variables are different:

- For binary: $x_i \in \{0,1\}$
- For integer: $x_i \in \mathbb{Z}^+$
- For continuous: $x_i \geq 0$

According to these constraints binary is the most restrictive one which allow for each items selected or not. Our python data shows the objective value is \$1761.

The integer model allows to use the same item again, so it is expected to increase the objective value. The result of objective function is \$3317 according to our code. This outcome has happened by items with high value to the weight volume ratio being selected multiple times to fill the remaining capacity.

The continuous model is giving the most freedom to the LP in which items can be taken in fractional amounts and no upper bound for the decision variables. In the result of the code mostly high value items at full quantity were chose. And the highest objective function value is \$3367.37.

In theoretical the expectation in maximization problems was $Z_{continuous} \geq Z_{binary}$. The increased domain enlarged the feasible region. The only constraint is $x_i \geq 0$. So, the continuous model's feasible region is also including binary and integer models. The integer model is more comprehensive than binary model. It allows multiple copies for each item. In the binary model is the most constrained that each item can be taken once.

Binary Result: 1761.00
Integer Result: 3317.00
Continuous Result: 3367.37

3.c-)

The dual formulation of the continuous knapsack model is:

Minimize

$$W = M y_1 + V y_2$$

Subject to:

$$m_j y_1 + v_j y_2 \text{ for all } j = 1, \dots, 50$$

$$y_1 \geq 0, y_2 \geq 0$$

where:

y_1 = shadow price of mass capacity

y_2 = shadow price of volume capacity

The optimal dual solution obtained from the model is:

$$y_1 = 1.237$$

$$y_2 = 1.431$$

Therefore,

$$W = 3367.373$$

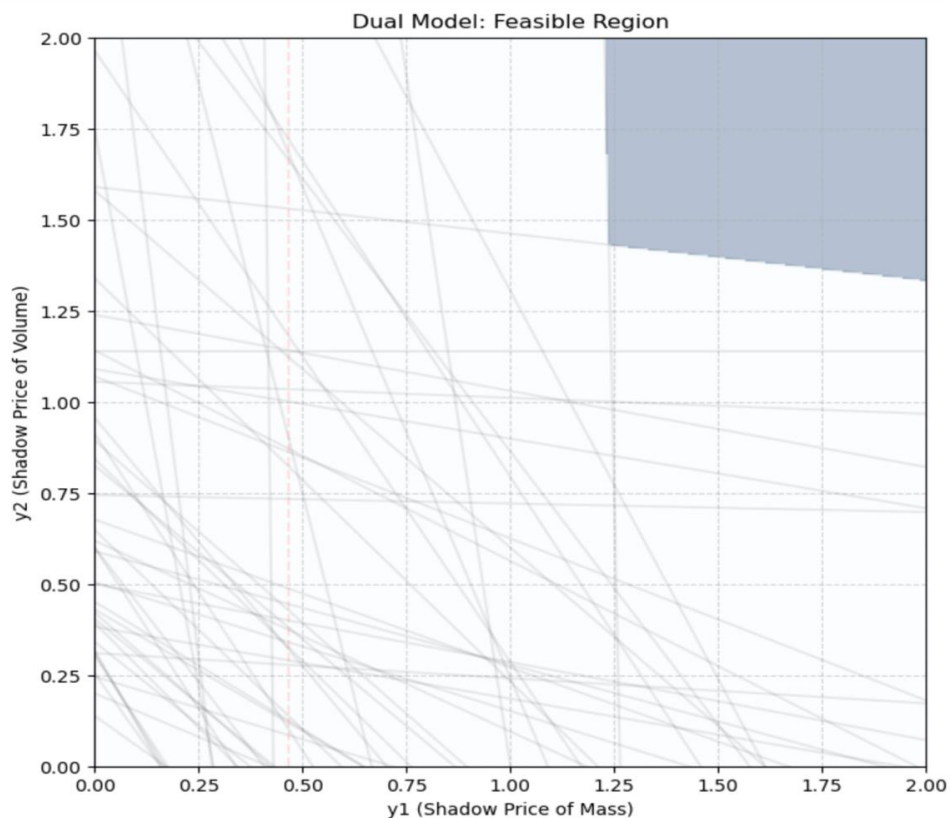
--- Result of 3c ---

Dual Objective (W): 3367.373

y1 (Mass Shadow Price): 1.237

y2 (Vol Shadow Price): 1.431

3.d-)



The shaded area represents the feasible region of the dual problem. Each line corresponds to one item constraint of the primal model. Since the dual is a minimization problem, the optimal solution is obtained at the lowest feasible corner point. The graphical solution is consistent with the numerical dual solution obtained in Part 3c.

3.e)

The shadow price is equal to the optimal dual variable value. The results are in the below:

Constraint	Shadow Price (Dual Variable)
Mass capacity ($\sum m_i x_i \leq 1262 kg$)	$y_1 = 1.237 \text{ \$/kg}$
Volume capacity ($\sum v_i x_i \leq 1262 m^3$)	$y_2 = 1.431 \text{ \$/m}^3$

y_1 and y_2 are the two primal constraints of dual LP. According to our results the dual objective value W is \$3367.373 which is same of the primal LP's objective value. It shows that strong duality is ensured.

The shadow price of a constraint represents the value effect on the one-unit change. One unit increase in the mass capacity would increase the optimal primal objective value is \$1.237. One unit increase in the volume capacity would result the optimal objective value \$1.431. Hence, the maximum amount payment for an additional unit of each resource:

- +1 kg mass capacity -> max \$1.237
- +1 m³ volume capacity -> max \$1.431

Since $y_2 > y_1$ the volume is more binding than mass constraint. At the optimal solution volume is the one that limits the objective value more. It is consistent with the primal optimal solution. At a higher marginal cost for volume means there were fewer items that can be trade value against volume compared to mass.

The shadow price tells us the maximum willingness to pay for one extra unit of capacity. If one additional cubic meter of volume costs less than \$1.431 it is worth. If one additional kilogram of mass costs less than \$1.237 it is worth buying. Since the volume shadow price is higher than mass, expanding volume capacity has more value than expanding mass capacity.

On the other hand, these values are only valid for small changes in each capacity independently, while other holds fixed. For large enough changes the optimal solution may shift.

AI Usage: AI based tools were used for proofreading and potential grammatical mistakes in the report. Additionally for the coding errors we used AI just to tell us the minor clarification. All mathematical formulations, data analysis, graphics, and python implementations are the original work of our group.